

TreeDataBase: Sensor-to-Analysis Data Pipeline

Input data

Climavi Sensor
(in-soil, 3 depths)
NB-IoT/LoRaWAN

QGIS Tree Database
(treeLocations.shp)
53 attributes/tree

DEM/DSM
(LiDAR, open data)
1 m resolution

Data management

Climavi Cloud
(ThingsBoard API)
Meteoblue weather

Manual digitisation
(QGIS polygons)
Land use at 3 radii

GIS processing
(QGIS Python scripts)
TWI · TPI · SVF

Processing

fetch_timeseries.py
→ sensor_data.csv
→ JSON snapshots

compute_landuse.py
→ land use fields
in treeLocations.shp

Analysis & Output

_poc_irrigation.py
Event detection
Feature extraction

urban_tree_report.py
Dataset statistics
PDF + JSON export

irrigation_events_all.csv
tree_data.json per tree
dataset_report.pdf

